PICK4ME ■■■

Hyperlocal Campus Marketplace & Crowdsourced Peer Commute Delivery Platform

EXECUTIVE PROJECT SUMMARY
Academic & Hackathon Presentation

Developer: Abhishek Kanadje
Live App: pick4me-l868.onrender.com

1. PROBLEM STATEMENT VS. PICK4ME SOLUTION

Campus & Hostel Pain Points (Existing System)	Pick4Me Innovations (Engineered Solution)
• **High Fees & Minimum Orders:** Apps like Blinkit/Swiggy charge ■30-■60 delivery/surge fees on small essentials (■10 pen, Xerox, lab kit). • **Gate Restrictions:** Outside delivery riders are blocked at main security gates; students walk late at night to fetch orders. • **Invisible Local Vendors:** Campus canteens, stationery and Xerox hubs lack an online catalog to showcase real-time stock. • **Unmonetized Daily Commutes:** Hundreds of students walk daily between gates and hostels with zero earnings opportunities.	• **Dynamic Micro-Fees (■20-■35):** Affordable, fair distance-based fees with zero minimum order quantity limits. • **Direct Doorstep Handover:** Fellow peer students deliver directly to hostel rooms, bypassing gate bans. • **Zero-Commission Storefronts:** Campus shops list their full inventory online, receiving instant sales without middlemen. • **Zero-Carbon Foot Traffic Deliveries:** Converts routine daily walking commutes into dignified micro-earnings (■200-■500/day).

2. 3-PARTY PEER LOGISTICS & ADVANCE ESCROW PROTOCOL

Zero Fraud Architecture: The buyer pays 100% upfront (Product Cost + Delivery Fee) via UPI QR into **Platform Escrow**. Funds are securely locked and disbursed in two automated verification stages:

Stage 1: Store Pickup Handover	Stage 2: Customer Handover & OTP Verification
• Peer commuter arrives at the campus merchant store. • Commuter inspects and collects the ordered items. • Shopkeeper confirms physical handover on their dashboard. • **Automatic Payout 1:** System instantly transfers 100% of the Product Amount directly to the **Shopkeeper's Wallet**.	• Commuter reaches buyer's hostel room. • Buyer physically verifies products and gives secret **4-digit Delivery OTP**. • Commuter inputs OTP in app for cryptographic verification. • **Automatic Payout 2:** System transfers Delivery Fee + **Speed Bonus** directly to **Commuter's Wallet**.

3. DYNAMIC DISTANCE SLA & SPEED BONUS ENGINE

Delivery Zone	Distance Range	Base Fee	Target SLA	Incentives & Rules
Within Campus	0.0 – 1.0 km	**■20.00**	25 Mins	Hostel-to-hostel or library delivery
Near Gate Market	1.0 – 2.5 km	**■35.00**	35 Mins	Main gate shops and student plaza
Outer Town Market	2.5 – 5.0+ km	**■55.00**	50 Mins	External city pharmacies & bookstores
Speed Bonus	Any Zone	**+■10.00**	Early Arrival	Auto-awarded for delivering before SLA window
Rush Delivery	Any Zone	**+■15.00**	Fast-Tracked	Priority surge placing order at top of tasks board

4. KEY PLATFORM INNOVATIONS & USER EXPERIENCE

■ P2P In-App Live Audio Calling	■ Verified Order Feedback & Star Rating
• Integrated in-browser calling modal powered by **HTML5 Web Audio API**. • Realistic dual-tone telephone ringing synthesizer (440Hz+480Hz). • Live active call timer (`00:15`), Mute microphone, Speaker, and End Call. • Seamless 1-click fallback to direct SIM Phone (`tel:`) and WhatsApp chat.	• Interactive 1 to 5-star rating widget enabled immediately upon delivery. • Allows customers to review speed, handling quality, and courier behavior. • Displays permanent ✓ **Verified Campus Delivery Feedback** badge. • Builds campus trust and driver reputational scoring.

■ Multi-Store Cart & Custom Errands	■ Commuter Wallet & Audit Ledger
• Amazon/Blinkit-style catalog across 5 campus store categories. • Live search, price sorting, real-time stock tags, and cart subtotals. • **Custom Errand Generator:** Request off-catalog items from any unlisted local shop with custom instructions and estimated budget.	• Real-time digital balance for commuters and store merchants. • Immutable transaction ledger with unique IDs (`#TXN-xxx`). • Automated tracking of product sales, delivery rewards, and speed bonuses. • Withdrawal capability linked to registered UPI VPA (`@upi`).

5. TECHNICAL ARCHITECTURE & COMPONENT STACK

Layer	Technology	Implementation Role
Backend Framework	**Python 3.11+, Flask 3.0**	Core MVC architecture, session handling, dynamic pricing & APIs
Cloud Database	**PostgreSQL (Neon.tech)**	24/7 Persistent Serverless Cloud Postgres (Zero data loss)
Local DB Fallback	**SQLite3 (`pick4me.db`)**	Local development engine with automatic relational schema adaptation
Frontend UI/UX	**HTML5, Modern CSS3, JS**	Custom responsive design system, Flexbox/Grid, zero bulky frameworks
Audio Synthesizer	**Web Audio API**	Client-side dual oscillator telephonic ringtone and call timer
Security & Auth	**PBKDF2:SHA256, 4-digit OTP**	Werkzeug cryptographic password hashing and two-factor delivery code
WSGI Server	**Gunicorn WSGI**	Production-grade WSGI web worker orchestration on Render Cloud

6. DATABASE SCHEMA & REPOSITORY STRUCTURE

6 Core Relational Tables: 1) users (id, name, email, phone, role, wallet_balance, is_verified); **2) shops** (id, owner_id, shop_name, category, address, phone, is_active); **3) products** (id, shop_id, name, category, price, in_stock); **4) cart_items** (id, user_id, product_id, quantity); **5) orders** (id, customer_id, commuter_id, shop_id, product_amount, delivery_fee, total_amount, pickup_otp, delivery_otp, status, feedback_rating, feedback_comment); **6) wallet_transactions** (id, user_id, order_id, amount, type, description).

Key Source Files: `app.py` (Core controller, APIs, calling & feedback routes); `database.py` (Dual Neon Postgres/SQLite adapter); `templates/order_details.html` (2-stage timeline, live calling modal & review form); `templates/admin_dashboard.html` (KYC control & catalog seeder); `static/css/style.css` (Bespoke styling & call animations); `Procfile` & `vercel.json` (Multi-cloud deployment descriptors).

7. IMPACT METRICS & LIVE ACCESS

■ 100% Carbon-Neutral Mobility	■ Dignified Student Micro-Earnings	■ Live Verified Deployment
Zero motorized fleet overhead. Converts natural student pedestrian and cycle foot traffic into a high-speed campus logistics network.	Enables commuting hostelers to earn ■200–■500 daily during regular routine transit without fixed hours or shifts.	**App:** https://pick4me-l868.onrender.com **Admin:** abhishekkanadje7@gmail.com **Pass:** Abhi*2007 (KYC & Escrow oversight)